CODE REVIEW

AI Agent Bug Review

Fashion Supply Chain Simulation

Seven verified defects affecting negotiation accuracy, state integrity, configurability, and fallback behavior.

Prepared November 3, 2026
Repository: URP-Full-Project

Executive summary

The review found seven defects in the AI negotiation agent. The most urgent failures can activate contract terms that the student did not accept: a revised discussion may leave an older offer active, disagreement can be misread as agreement, and incomplete model output can silently supply missing values.

Four findings were reproduced with controlled model responses. Three additional findings were confirmed through code inspection and were already recorded in the repository's broader bug review. No live AI-provider calls were required.

PRIORITY	FINDING	PRIMARY RISK
High	Stale offer remains active after revised terms	Student accepts terms different from the visible discussion
High	Disagreement can trigger finalization	Contract offer created without affirmative consent
High	Incomplete responses invent missing terms	Unagreed values become executable contract terms
Medium	Supplier personality is ignored	Configured agents behave identically
Medium	Wrong demand history reaches the AI	Advice conflicts with the student's scenario
Medium	Instructor prompts are ignored	Configuration UI promises changes that never occur
Medium	No-AI negotiation reaches a dead end	Players cannot recover after a rejected proposal

Recommended order

Fix offer state and agreement validation first, then require complete contract data. After those integrity fixes, connect personality, scenario history, and instructor configuration to the prompt. Finally, provide a deterministic fallback path when no provider is configured.

Detailed findings

1 Revised terms can leave the old offer active — HIGH

IMPACT	After the AI has produced a draft, later chat cannot replace it. The supplier can announce revised terms while the accept action still activates the previous contract.
EVIDENCE	A controlled route reproduction began with a $30 wholesale draft. The next supplier message announced a revised $25 wholesale price but returned no new draft. Accepting the offer activated the original $30 contract.
LOCATION	backend/app/services/ai_service.py:366; backend/app/routes/negotiation.py:269
FIX	Treat every supplier message that changes contract terms as an offer-state update. Clear or replace the stored draft when negotiations resume, and bind acceptance to an explicit offer identifier or version shown to the student.

2 Disagreement can trigger finalization — HIGH

IMPACT	The phrase matching logic treats negative statements such as 'I do not agree' as possible agreement because it searches for substrings. The parsed response can create an offer even when the model reports that negotiation is incomplete.
EVIDENCE	A controlled response confirmed that 'I do not agree' appended the finalization instruction. A JSON response with negotiation_complete set to false still created a draft contract because that field is never checked.
LOCATION	backend/app/services/ai_service.py:349-366 and 454-460
FIX	Use intent-aware agreement detection with negation handling and word boundaries. Require negotiation_complete to be exactly true before reading contract data or creating a draft.

3 Incomplete AI responses invent contract terms — HIGH

IMPACT	Missing values are replaced with executable defaults. A partial response can therefore create a contract containing terms that neither party discussed.
EVIDENCE	A response containing only wholesale_price=30 produced a draft with buyback_price=0 and the configured minimum length of 1. Those values passed structural validation.
LOCATION	backend/app/services/ai_service.py:493-506
FIX	Require all mandatory fields, reject null or missing values, and validate them against the explicit terms in the conversation. Do not substitute business terms with defaults during agreement finalization.

Detailed findings (continued)

4 Supplier personality has no effect when AI is available — MEDIUM

IMPACT	The game assigns personalities, but the live AI prompts omit that value. Different supplier personalities therefore receive identical instructions and can behave identically.
EVIDENCE	Captured provider calls for selfish and altruistic states were identical in both proposal evaluation and chat. Personality affects only the simple fallback evaluator.
LOCATION	backend/app/services/negotiation_service.py:86; backend/app/services/ai_service.py:308
FIX	Add a clear personality objective and decision policy to both prompts. Include behavioral tests that capture provider messages for each personality and verify meaningful differences.

5 The AI uses the wrong demand history — MEDIUM

IMPACT	Negotiation uses the global default history instead of the active session's selected scenario. Supplier reasoning can conflict with the demand data visible to the player.
EVIDENCE	A session populated with demand 99,999 did not pass that value to the captured prompt. The prompt instead reported statistics from the global history.
LOCATION	backend/app/services/ai_service.py:273; backend/app/services/negotiation_service.py:83
FIX	Pass the session's historical_demands into proposal evaluation and chat generation. Keep global history only as a deliberate fallback when no game state exists.

6 Instructor prompt changes have no effect — MEDIUM

IMPACT	The application saves a system prompt template and example dialogue, but neither negotiation path includes them in provider messages.
EVIDENCE	A distinctive custom instruction loaded through negotiation configuration was absent from captured chat and proposal-evaluation prompts.
LOCATION	backend/app/services/ai_service.py:295-308; backend/app/services/negotiation_service.py:86
FIX	Compose the configured template and examples into both prompt paths while keeping non-negotiable output and validation rules in a separate application-owned instruction block.

7 No-AI negotiations reach a dead end — MEDIUM

IMPACT	When no provider is configured, fallback chat only returns generic text and never creates an offer. After an initial rejection, the proposal form is hidden and ordering remains unavailable.
EVIDENCE	Controlled fallback chat with complete revised terms returned no draft. Frontend visibility logic hides the proposal form during the ongoing negotiation.
LOCATION	backend/app/services/ai_service.py:264-269; frontend/main.js:860-863
FIX	Implement deterministic term extraction and evaluation for fallback chat, or restore the proposal form so the student can submit a revised structured proposal.

Validation notes

The controlled checks executed the relevant service and route functions with mocked provider responses. They did not call OpenAI, OpenRouter, or DeepSeek. This made the results deterministic and isolated application behavior from model variability.

Confirmed reproductions

1. Explicit disagreement triggered the finalization prompt and negotiation_complete=false still produced a draft.
2. A partial contract response supplied buyback price 0 and length 1.
3. A revised $25 discussion left the previous $30 offer active and accept activated $30.
4. Selfish and altruistic suppliers produced identical provider prompts.
5. Session-specific demand and instructor configuration were absent from captured prompts.
6. With no provider, revised terms could not create an offer.

Scope and limitations

This report focuses on the AI negotiation agent and its immediate state transitions. It does not claim that every possible provider response or end-to-end browser path was tested. Findings 5-7 overlap with the repository's existing broader bug review and were rechecked against the current implementation.

Acceptance criteria for the fix

A fix should ensure that the offer shown to the student is exactly the offer accepted; negative or ambiguous language never finalizes a contract; every finalized contract contains complete, explicitly agreed terms; personality, scenario history, and instructor settings reach both AI paths; and negotiation remains completable without an external AI provider.